SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	*Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)*		
Student Name:	G. Raju	Reg. No.:	192411065
Section / Batch:	CSE (2024)	Date:	

Instructions to Students

- Answer BOTH scenario-based questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type		Questions	Marks
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1 - Q2	100 (2 × 50)
	GRAND TOTAL:		100 Marks

SECTION A – SCENARIO BASED ASSESSMENT

Code of Conduct

I G. Raju (192411067) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G. Raju

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Question No.	Problem Understanding & Context Analysis (25%)	Application of Concepts (30%)	Analytical & Decision-Making Skills (20%)	Solution Design & Approach (15%)	Clarity, Justification & Presentation (10%)	Total Marks (Each – 50)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________	Signature: ________	Signature: ________
Date: ________	Date: ________	Date: ________

Course Name :- ~~CSA0648~~ DAA

Course code :- CSA0618

Name :- G. Raju

reg :- 192411065

Co2- AT2

1. Linear Search

(a) explain how linear search operator as this scenario

linear search checks each transaction ID one by one from the begining of the unsorted list until the required transaction is found an the entire list has been searched

(b) Analyze the time complexity for best Average & Worst Case :-

Case	Time Capacity	Reason
Best case	O(1)	transaction id is found at the 1st position.
Average case	O(n)	About half of the records are checked
Worst Case	O(n)	transaction is last or not present.

(c) Suggest whether Scarting + binary seares would Compare performance & festity

Sorting the data & using binary search reduce searching time to O(logn) However, because How records are inserted frequently, maintaining sorted order requires repeated sorting & shifting element.

∴ linear search is suitable for frequently changing data.

2. Brute force - farthest pair of point

(a) Explain how the brute - force method works.

The Algorithm Calculate the distance b/w every possible pair of delivery points & Compares the distances the pair with the Maximum distance is selected as the farthest pair.

(b) Derive the time Complexity

for n points :

No. of pairs, $\frac{n(n-1)}{2}$

Since every pair is checked

$T(n) = O(n^2)$

Analyze why this approach becames inefficient

As the no. of points increases the no. of comparsion quadractically for example.

- 100 points → 4950 comparision
- 1000 points → 499500 comparsion

This greatly increases processing time making brute force unsuitable for large data sets.

8. Brute force string matching

(a) explain how brute - force string matching insolcs

The Algorithm compares the keyword with every possible in the document.

if character match it Continues otherwise, it shifts the keywords by one position.

(b) Analyzer the time Complexity

(i) Let :-

- text length = n
- Pattern length = m

(ii) Worst case complexity : o(nm)

(iii) Best case : o(n)

© Suggest why it is inefficent & justify optimized algorithm.

· Brute force matching performas many repeated character comparison making it slow for large documents Algorithm like KMP & - more avoid unnecessary comparison reducing execution time & improving efficiency for large scale text searching

9. Binary Search :-

ⓐ explain how Binary search works.

Binary search works only on sorted data. it compare the target mark with the middle element

- if equal → record found
- if smaller → search the left half
- if larger - search the right half

Analyzer the time complexity

Case	time complexity	Reason
Best case	o(1)	middle element is the target.
Average case	o(logn)	search space is naked
worst case	o(logn)	Continues until one element remains

© Discuss why maintaining sorted order is critical.

. Binary search depends in sorted data random insertion distrub the order causing incorrects search result.

∴ The dataset must to sorted after updates maintaining sorted order ensures efficient o(logn) searching although insertion may require o(n) time.

10. Brute force geometric Andysis

a) explain how brute force geometric works

- The Algorithm checks every possible Combination of obstacle / Convex hell)
- It tests all candidate edges & vertics whether the remaining points the each edge.

B) Andyzer the time complexity for n points

- All points pairs are examined
- each pair is checked against add remaining points

$O(n^3)$

c) The Cubic time Complexity make braute-force geomatric Andysis.

Impractical for large data sets because as the no. of computation increase rapidly points glow efficient. algorithm such as gracham scan or jarvis march compute.